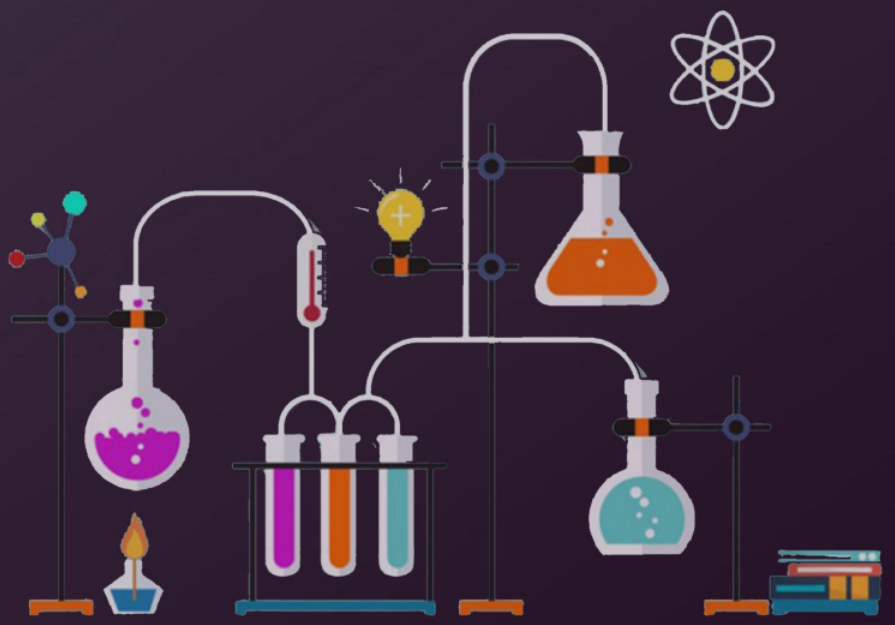
CHEMISTRY

Salt Analysis



SHEET

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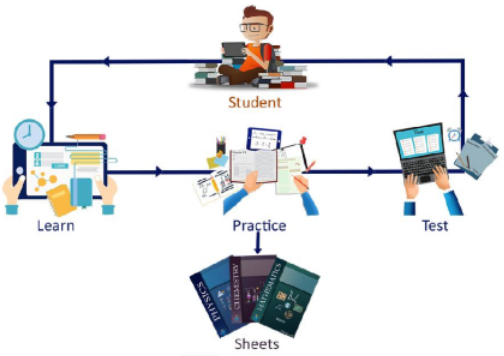
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Types of Reactions	1 - 2	22	1 - 2	
2	Thermal Decomposition Reaction				
3	Test of Carbonate and Bicarbonate Ions	3	23	3	16 - 17
4	Test of Sulphite Ion	4 - 5			
5	Test Of Thiosulphate Ions				
6	Test of Sulphide Ions	6			
7	Test of Nitrite	7		4	
8	Test of Acetate				
9	Testing of Fluoride and Chloride Ions		1	5	
10	Test of Bromide Ion				
11	Test of Iodide Ion			6	
12	Test of Nitrate and Oxalate Ions			7 - 8	1
13	Test of Sulphate and Phosphate Ions	8	24	9 - 11	19
14	Test of Chromate, Dichromate and Permanganate	9 - 14	2, 25 - 26	12 - 13	2, 7 - 9, 20
15	Dry Tests for Cations	15			
16	Classification of Cations				
17	Test for Cations Of Group 1				
18	Test of Mercury(II) and Copper Ions	16	27	14	3, 21
19	Test of Bismuth and Cadmium Ions	17 - 19	3, 9 - 11, 28	15	4 - 5
20	Test of Cations of Group II-B	20 - 21	12 - 14, 18	16 - 18	
21	Test for Iron(III) Cation of Group III	22 - 23	4, 19	19 - 23	
22	Test of Aluminium(III) and Chromium(III) Ions	24	5		
23	Test of Group IV Ions	25			
24	Test of Group V Cations	26 -28	6		
25	Test of Group VI and Group Zero Cations	29 - 30	15 - 17	24	10 - 12
26	ACC	31 - 37	7 - 8, 20 - 21, 29 - 30	25 - 34	6, 13 - 15, 18, 22 - 24

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Types of Reactions

Q1. H_2S and SO_2 gas can be distinguished by –
(A) litmus paper
(B) lime water
(C) lead acetate paper
(D) HCl

Q2. When H_2S gas is passed through the aqueous solution of a salt a white precipitate is formed. The solution contains ions of –
(A) Pb (B) Zn (C) Cu (D) Ni

Test of Carbonate and Bicarbonate Ions

Q3. Soda extract of the salt is prepared by :
(A) fusing mixture of backing soda and slat and then extracting with water
(B) dissolving $NaHCO_3$ and salt in dil HCl
(C) boiling Na_2CO_3 and salt in dil HCl
(D) boiling Na_2CO_3 and salt in distilled water

Test of Sulphite Ion

Q4. $[X] + H_2SO_4 \longrightarrow [Y]$ a colourless gas with suffocating smell
$[Y] + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ green solution
[X] and [Y] is –
(A) SO_3^{2-}, SO_2 (B) Cl^-, HCl
(C) S^{2-}, H_2S (D) CO_3^{2-}, CO_2

Q5. Hydrogen sulphide in amonical medium is a group reagent for :
(A) 2nd group radicals
(B) group II radicals
(C) group IV radicals
(D) group V radicals

Test of Sulphide Ions

Q6. If HCl is not added before passing H_2S in the second group, it may result in the –
(A) incomplete precipitation of second group sulphides
(B) precipitation of sulphides of cation belonging to the subsequent groups
(C) precipitation of sulphur
(D) precipitation of lead as lead sulphide

Test of Nitrite

Q7. When concentrated H_2SO_4 is added to dry KNO_3, brown fumes are evolved. These fumes are due to :
(A) SO_2 (B) $SO_2 + SO_3$
(C) NO (D) NO_2

Test of Sulphate and Phosphate Ions

Q8. Which of the following give(s) canary yellow ppt. with ammonium molybdate?
(A) PO_4^{3-}
(B) As^{3+}
(C) Both (A) and (B)
(D) None of these

Test of Chromate, Dichromate and Permanganate

Q9. A pale green crystalline inorganic salt (A) dissolve freely in water. It gives brown precipitate on addition of aqueous solution NaOH. The solution of (A) also gives a black precipitate on bubbling H_2S in alkaline medium. An aqueous solution of (A) decolourized the pink colour of the permanganate solution. The metal in the salt solution is :
(A) copper (B) aluminium
(C) lead (D) iron

Q10. Potassium chromate solution is added to an aqueous solution of a metal chloride. The precipitate thus obtained are insoluble in acidic acid. These are subjected to flame test, the colour of the flame is :

(A) Lilac (B) apple green

(C) crimson red (D) golden yellow

Q11. Consider the following observation ,

M^{n+} + HCl $\longrightarrow$ white precipitate $\xrightarrow{\Delta}$ water solution.

The metal ion M^{n+} will be :

(A) Hg^{2+} (B) Ag^{+}

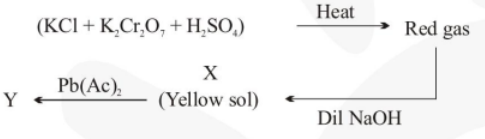
(C) Pb^{2+} (D) Sn^{2+}

Q12. $K_2Cr_2O_7$ gives blue colour with H_2O_2 & H_2SO_4 the blue colour will convert into green colour. Blue & green compounds are –

(A) CrO_5 & Cr_2O_3 (B) $Cr_2O_7^{2-}$ & CrO_4^{2-}

(C) CrO_5 & Cr (D) All the above

Q13.



The formula and colour of X are respectively –

(A) CrO_2Cl_2, red (B) Pb CrO_4, yellow

(C) Na_2CrO_4, yellow (D) $Cr_2(SO_4)$, green

Q14. A solution of $Na_2C_2O_4$, when added to a solution containing Ba^{2+}, Sr^{2+} and Ca^{2+} ions, will precipitate.

(A) Ca^{2+} ions

(B) Ca^{2+} and Sn^{2+} ions

(C) Ba^{2+}, Sr^{2+} and Ca^{2+} ions

(D) none of these

Dry Tests for Cations

Q15. Which of the following salt is used in borax bead test ?

(A) $K_2CO_4Al_2(SO_4)_3 24H_2O$

(B) $Na_2B_4O_7.10H_2O$

(C) $CaSO_4.2H_2O$

(D) none of these

Test of Mercury(II) and Copper Ions

Q16. Hypophosphorus acid when it combine with $CuSO_4$ solution will get red coloured compound X. What is X ?

(A) Cu_2O (B) Cu_2H_2

(C) CuO (D) Cu_2S

Test of Bismuth and Cadmium Ions

Q17. Which of the following pairs of ions cannot be separated by H_2S in dil. HCl?

(A) Bi^{3+}, Sn^{4+} (B) Al^{3+}, Hg^{2+}

(C) Zn^{2+}, Cu^{2+} (D) Ni^{2+}, Cu^{2+}

Q18. Yellow ammonium sulphide solution can be used for the separation of which of the following pair of species ?

(A) CuS and PbS (B) PbS and Bi_2S_3

(C) Bi_2S_3 and CuS (D) CdS and As_2S_3

Q19. Aqueous solution of

(M)+$(NH_4)_2S$→yellow ppt(B) $\xrightarrow{(NH_4)_2S_2}$ insoluble.

The cation present in (M) is –

(A) CdS (B) SnS_2

(C) Cd^{2+} (D) Sn^{2+}

Test of Cations of Group II-B

Q20. Addition of solution containing $C_2O_4^{2-}$ ions to an aqueous solution containing Ba^{2+}, Sr^{2+} and Ca^{2+} will precipitate :

(A) Ca^{2+} (B) Ca^{2+} and Sr^{2+}

(C) Ba^{2+} and Sr^{2+} (D) all three

Q21. (Yellow ppt.) T $\xleftarrow{K_2CrO_4}$ X $\xrightarrow{\text{dilute HCl}}$ Y (Yellow ppt.) + Z↑ (pungent smelling gas).

If X gives green flame test , then X is :

(A) $MgSO_4$ (B) BaS_2O_3

(C) $CuSO_4$ (D) PbS_2O_3

Test for Iron(III) Cation of Group III

Q22. Formation of a rose-red precipitate when a slightly alkaline solution of an inorganic salt is treated with dimethyl glyoxime confirms the presence of –

(A) cobalt (B) zinc
(C) iron (D) nickel

Q23. Certain yellow coloured solid gives reddish brown precipitate in group III of qualitative analysis. The solid does not react with dilute H_2SO_4 but when few drops of $KMnO_4$ solution is added to the hot suspension of salt in dilute H_2SO_4, its pink colour is discharged with effervescence and evolution of CO_2. the slat is likely to be –

(A) iron (II) oxalate (B) ferrous sulphate
(C) ferric carbonate (D) ferric chloride

Test of Aluminium(III) and Chromium(III) Ions

Q24. An inorganic salt in its aqueous solution produced a white ppt. with NaOH which dissolves in excess of NaOH. Also its aqueous solution produced light yellow ppt., with $AgNO_3$ sparingly soluble in NH_4OH. The probable salt is –

(A) $AlBr_3$ (B) AlI_3
(C) $AlCl_3$ (D) $ZnCl_2$

Test of Group IV Ions

Q25. A compound (A) forms a unstable pale blue colour solution in water. The solution decolourised Br_2^- water and an acidified solution of $KMnO_4$. The possible compound (A) is:

(A) HNO_2 (B) HNO_3
(C) N_2O_5 (D) None of these

Test of Group V Cations

Q26. When the solution of an inorganic salt in sodium hydroxide is boiled, a pungent gas is produced, which turns moist red litmus paper blue and a mercurous nitrate solution becomes black. This indicates the presence of ..... ion in the salt.

(A) CO_3^{2-} (B) NH_4^+
(C) Sn^{4+} (D) NO_2^-

Q27. In the precipitation of the radicals of iron group in qualitative analysis, NH_4Cl is added before adding NH_4OH. This causes :

(A) decrease in the concentration of OH^- ions
(B) removal of PO_4^{3-} ions
(C) increase in the concentration of Cl^- ions
(D) increase in the concentration of NH_4^+ ions

Q28. Magnesium carbonate does not precipitate with the carbonates of group V radicals in presence of NH_4OH and NH_4Cl because :-

(A) $MgCO_3$ is soluble in water
(B) $MgCO_3$ is soluble in NH_4OH
(C) $MgCO_3$ is soluble in NH_4Cl
(D) $MgCO_3$ is soluble in $(NH_4)_2CO_3$

Test of Group VI and Group Zero Cations

Q29. A metal X on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through $CuSO_4$ solution gives a blue colour. Y is –

(A) $Mg(NO_3)_2$ (B) Mg_3N_2
(C) NH_3 (D) MgO

Q30. To an aqueous solution containing Hg_2^{2+}, Hg^{2+}, Pb^{2+} and Cd^{2+}. (dil.) HCl ($\approx$ 6 M) is added. This causes the precipitation of :

(A) Hg_2Cl_2 only
(B) Hg_2Cl_2 and $PbCl_2$
(C) $PbCl_2$ only
(D) $PbCl_2$ and $HgCl_2$

ACC

Q31. Heating $(NH_4)_2Cr_2O_7$ yields a gas, which is also obtained by –

(A) Mg_3N_2 in H_2O
(B) heating NH_4NO_2
(C) heating NH_4NO_3
(D) Na_2O_2 in H_2O

Q32. Which of the following ion is detected by Nessler's reagent ?

(A) MnO_4^- (B) NH_4^+

(C) CrO_4^{2-} (D) PO_4^{3-}

Q33. Three separate samples of aqueous solution of X gave following results. One formed white ppt. with excess of ammonia solution, one formed white precipitate with dil NaCl solution and one formed a black precipitate with H_2S. the salt could be –

(A) $AgNO_3$ (B) $Pb(NO_3)_2$

(C) $Hg(NO_3)_2$ (D) $MnSO_4$

Q34. When a reagent (A) react with Fe^{3+} the solution turns red due to the formation of a compound (B). The reagent causes no change in colour when it reacts with Fe^{2+} in the pure state. (A) and (B) are respectively –

(A) $K_4[Fe(CN)_6]$ and $Fe_4[Fe(CN)_6]_3$

(B) NH_4SCN and $[Fe(SCN)_3]$

(C) $K_3[Fe(CN)_6]$ and $K_2Fe[Fe(CN)_6]$

(D) Na_2HPO_4 and $FePO_4$

Q35. Na_2CO_3 cannot be used in place of $(NH_4)_2CO_3$ for the precipitation of fifth group radicals, since:

(A) magnesium will be precipitated

(B) concentration of carbonate ions is very low

(C) sodium ions will react with acidic radicals

(D) Na^+ ions interfere with the detection of fifth group

Q36. An aqueous solution of $FeSO_4$, $Al_2(SO_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The materials obtained are :

(A) a colouless filtrate and a green residue

(B) a yellow filtrate and a green residue

(C) a yellow filtrate and a brown residue

(D) a green filtrate and a brown residue

Q37. Colourless solutions of the following four salts are placed separately in four different test tubes and a strip of copper is dipped in each one of these. Which solution will turn blue ?

(A) KNO_3 (B) $AgNO_3$

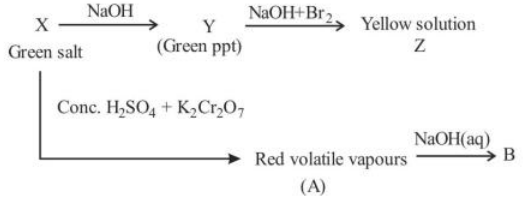
(C) $Zn(NO_3)_2$ (D) $ZnSO_4$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct Joption(s) to answer the question.*

Q1. Which of the following statements is/are true?



(A) The yellow solution z is same as B

(B) The compound Y is $Cr(OH)_3$

(C) The compound A is CrO_2Cl_2

(D) The salt X is a chloride

Q2. Which of the following statement(s) is/ are correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with concentrated H_2SO_4 ?

(A) A deep red vapour is evolved

(B) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4

(C) Chlorine gas is evolved

(D) Chromyl chloride is formed

Q3. Brown ring test for nitrate fails if the mixture of salts, contain along with nitrate, the :

(A) NO_2^- ions (B) CO_3^{2-} ion

(C) Br^- ions (D) SO_4^{2-} ions

Q4. $K_4[Fe(CN)_6]$ is used in the detection of :

(A) Fe^{2+} (B) Fe^{3+}

(C) Cu^{2+} (D) Zn^{2+}

Q5. What is/are correct about $Al_2(SO_4)_3$?
(A) It is a white salt.
(B) Its aqueous solution gives precipitate of compound Al with $AgNO_3$ solution.
(C) On adding NaOH solution to its aqueous solution, a precipitate is formed which dissolved in excess of NaOH.
(D) It will give precipitate with lead acetate solution.

Q6. The reagents, NH_4Cl and aqueous NH_3 will precipitate :
(A) Ca^{2+} (B) Al^{3+} (C) Ba^{2+} (D) Mg^{2+}

Q7. Na_2CO_3 cannot be used in place of $(NH_4)_2CO_3$ for the precipitation of fifth group radicals. Choose the incorrect statements –
(A) magnesium will be precipitated
(B) concentration of carbonate ions is very low
(C) sodium ions will react with acidic radicals
(D) Na^+ ions interfere with the detection of fifth group

Q8. Which of the following sulphate is soluble in water ?
(A) $CuSO_4$ (B) $CdSO_4$
(C) $PbSO_4$ (D) $Bi_2(SO_4)_3$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 9 to 11

The yellow coloured precipitate of compound (A) is formed on passing H_2S through a neutral solution of salt (B). Compound (A) is soluble in hot dilute HNO_3 but insoluble in yellow ammonium sulphide . The solution of (B) on treatment with small quantity of aq. NH_3 gives white precipitate which becomes soluble in excess of it forming a compound (C). The solution of (B) gives white precipitate with small concentration of KCN which becomes soluble in excess of this reagent forming a compound (D). The solution of (D) on treatment with H_2S gives (A), The solution of $BaCl_2$ gives white precipitate of compound (E) with 'B' which is insoluble in conc. HNO_3.

Q9. What is compound A
(A) CdS (B) $CdSO_4$
(C) SnS_2 (D) $CrCl_3$

Q10. The hybridisation in compound C is:
(A) dsp^2 (B) sp^3
(C) d^2sp^3 (D) sp^3d^2

Q11. The compound E is insoluble in water because:
(A) Lattice energy is greater than hydration energy
(B) Lattice energy is less than hydration energy
(C) the presence of polar bond in compound E
(D) it can't predicted

Paragraph for Q. 12 to 14

Using dil. H_2SO_4 gas evolution takes place from several acid radicals. If we consider their smell, colour and other observation if any, several acid radicals can be detected in this step.

Q12. Using dil. H_2SO_4, which of the acid radical cannot be confirmed.
(A) NO_2^- (B) $CH_3CO_2^-$
(C) HSO_3^- (D) S^{2-}

Q13. When the aqueous suspension of Ag_2SO_3 and Ag_2CO_3 are heated, the respectivel ppts. are obtained as –
(A) Ag_2O and Ag_2O
(B) Ag_2O and Ag
(C) Ag and Ag_2O
(D) Ag and Ag

Q14. Which of the following radical(s) is/are producing same gas on treated with (Zn + dil. H_2SO_4)
I. SO_3^{2-} II. HSO_3^-
III. S^{2-} IV. Cl^-
(A) I and II only
(B) I, II and III only
(C) I, II, III and IV
(D) II, III and IV only

Paragraph for Q. 15 to 17

A is a colourless crystalline salt which is soluble in water to form a super saturated solution. Salt (A) reacts with salt (B) (which is sensitive to light & gets photo reduction to give its metal) it gives a white ppt (C) and with excess of salt (A), (C) gives a soluble complex (D). (C) in exposure in air and (D) on heating both produces same black coloured ppt (E). (A) is used in medicine and also in extraction of gold and decoloured I_2 in KI solution.

Q15. When salt (A) reacts with HCl solution it decompose to give
(A) brown coloured gas
(B) rotten eggs smelled gas
(C) white turbidity
(D) violet coloured gas

Q16. When salt (A) is heated to 223°C the product obtained are
(A) Na_2SO_4, Na_2S_5
(B) PbO, SO_2
(C) CuO, SO_2
(D) $FeCl_2$, Cl_2

Q17. Identify (E)
(A) Hg_2O
(B) Ag_2S
(C) PbO_2
(D) CuS

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q18. Match the column –

Column-I (sulphlide)	**Column-II (colour/group)**
(A) HgS	(P) 2nd group of basic radical
(B) NiS	(Q) Buff colour
(C) MnS	(R) Dirty white
(D) ZnS	(S) Black

Q19. Match the column

Column I	**Column II**
(A) Fe^{2+} ion	(P) Aluminon reagent
(B) Ni^{2+} ion	(Q) Alizarin reagent
(C) Co^{2+} ion	(R) Ammonium thiocyanate reagent
(D) Al^{3+} ion	(S) Dimethyl glyoxime reagent

Q20. Match the observation of Column I with inferences in column II

Column I	**Column II**
(A) Canary yellow precipitate with ammonium molybdate	(P) NO_3^-
(B) Brown ring test with dil. H_2SO_4	(Q) SO_4^{2-}
(C) Yellop ppt. with $HgCl_2$ solution	(R) PO_4^{3-}
	(S) NO_2^-

Q21. Match the column –

Column I [Characteristc of salt]	**Column II**
(A) Leave(s) no residue on heating	(p) $Ag_2C_2O_4$
(B) Form(s) brown ring on reaction with $FeSO_4$ and Conc. H_2SO_4	(q) $(NH_4)_2CO_3$
(C) Colourless acidic gas is evolved on heating	(r) $Pb(NO_3)_2$
	(s) NH_4NO_3

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q22. **Statement 1 :** In $ZnCl_2$ solution H_2S gas is passed, No ppt. is obtained.

Statement 2 : H_2S is weak acid and does not dissociate completely.

Q23. **Statement - 1 :** CO_2 is a acidic oxide.

Statement - 2 : The reaction of CO_2 with water produces H_2CO_3.

Q24. **Statement - 1 :** Sulphates are usually estimated as $BaSO_4$ and not as $MgSO_4$

Statement - 2 : $MgSO_4$ water soluble while $BaSO_4$ in soluble

Q25. **Statement - 1 :** $KMnO_4$ is coloured.

Statement - 2 : The colour of a compound appears due to charge transfer and electronic transition.

Q26. **Statement - 1 :** $Na_2Cr_2O_7$ is not a primary standard in volumetric analysis.

Statement - 2 : $Na_2Cr_2O_7$ is hygroscopic.

Q27. **Statement - 1 :** The species $[CuCl_4]^{2-}$ exists while $[CuI_4]^{2-}$ does not.

Statement - 2 : I^- is stronger reducing agent then Cl^-.

Q28. **Statement - 1 :** A very dilute acidic solution of Cd^{2+} and Ni^{2+} gives yellow precipitate of CdS on passing hydrogen sulphide.

Statement - 2 : Solubility product of CdS is more than that of NiS.

Q29. **Statement - 1 :** AgCl dissolves in NH_4OH.

Statement - 2 : AgCl reacts with NH_4OH and forms AgOH, which is soluble in water.

Q30. **Statement - 1 :** On addition of $FeCl_3$ solution in NH_4SCN solution become deep red colour.

Statement - 2 : Red coloration developed due to formation of $Fe(SCN)_3$ complex.

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. An inorganic salt, when treated with conc. H_2SO_4 produced oily drops. The probable salt is -

(A) an oxalate (B) a fluoride
(C) a nitrite (D) an iodide

Q2. Iron becomes passive by _____ due to formation of _____ .

(A) dilute HCl , Fe_2O_3
(B) conc. HNO_3 , Fe_3O_4
(C) conc. H_2SO_4 , Fe_3O_4
(D) conc. HCl , Fe_3O_4

Q3. A mixture of two salts is not soluble in water but dissolves completely in dil HCl to form a colourless solution. The mixture could be -

(A) $AgNO_3 + KBr$
(B) $BaCO_3 + ZnS$
(C) $FeCl_3 + CaCO_3$
(D) $Mn(NO_3)_2 + MgSO_4$

Q4. A compound produced blue mass in cobalt nitrate test and formed a brown ring with $FeSO_4$ and conc.H_2SO_4. The probable compound is:

(A) $Al(NO_3)_3$ (B) $Zn(NO_3)_2$
(C) $Cu(NO_3)_2$ (D) $MgCl_2$

Q5. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with the evolution of a colourless gas 'Y'. Identify 'X' and 'Y'.

(A) $X = CO_2, Y = Cl_2$
(B) $X = Cl_2, Y = CO_2$
(C) $X = Cl_2, Y = H_2$
(D) $X_2 = H_2, Y = Cl_2$

Q6. $X + NH_3 + KOH \longrightarrow$ Y (Brown precipitate) . X and Y are respectively :

(A) Nessler's reagent , iodide of Million's base
(B) Iodide of Million's base , Nessler's reagent
(C) Iodide of Million's base , prussian blue
(D) Nessler's reagent , golden spangles

Q7. Lead (II) sulphate is soluble in –

(A) conc. HNO_3
(B) conc. HCl
(C) water
(D) ammonium acetate

Q8. Which of the following gives black precipitate on passing H_2S through it –

(A) acidified zinc nitrate solution
(B) ammoniacal barium chloride solution
(C) magnesium nitrate solution
(D) copper nitrate solution

Q9. An aqueous solution contains SO_4^{2-} and Cl^- ions. Which of the following aqueous solution will precipitate only SO_4^{2-} ion but not Cl^- ion ?

(A) NaOH (B) KOH
(C) $BaCl_2$ (D) $BaSO_4$

Q10. The aqueous solution of which of the following reagent will give Prussian blue coloured ppt. with an aqueous solution containing iron (III) ions :

(A) potassium thiocyanate
(B) potassium hexacyaoferrate (II)
(C) potassium pyroatimonate
(D) all of these

Q11. Which pair of ions would be expected to form precipitated when their dilute aqueous solutions are mixed :

(A) Na^+, SO_3^{2-} (B) NH_4^+, CO_3^{2-}
(C) Na^+, S^{2-} (D) Fe^{3+}, PO_4^{3-}

Q12. On adding a solution of CrO_4^{2-} ions to an aqueous solution containing Ba^{2+}, Sr^{2+} and Ca^{2+} ions. The precipitate obtained first of all will be :

(A) $CaCrO_4$

(B) $SrCrO_4$

(C) $BaCrO_4$

(D) a mixture of all the three

Q13. Ammonium dichromate is used in some fireworks. The green coloured powder blown in air is due to :

(A) CrO_3 (B) Cr_2O_3

(C) Cr (D) $CrO(O_2)$

Q14. The constituents of which of the following pairs of ions in a dilute HCl medium cannot be separated by passing H_2S gas through it ?

(A) Ca^{2+} and Hg_2^{2+} (B) Cu^{2+} and Cd^{2+}

(C) Zn^{2+} and Sn^{2+} (D) Co^{2+} and Cu^{2+}

Q15. Which of the following reagent will separate Bi^{3+} in one step from mixture having Fe^{3+}, Bi^{3+}, Cd^{2+}-

(A) dil. HCl + H_2S

(B) Excess NH_4OH

(C) Excess NaOH

(D) KI

Q16. A mixture of zinc oxide and cupric oxide can be separated by –

(A) boiling the mixture with caustic potash solution and filtering

(B) boiling the mixture with dilute HNO_3 and filtering

(C) heating with carbon

(D) removing cupric oxide with magnet

Q17. The reagent that can distinguish between silver and lead salt is –

(A) H_2S gas

(B) Hot dilute HCl solution

(C) NH_4Cl (solid) + NH_4OH (solution)

(D) NH_4Cl (solid) + $(NH_4)_2CO_3$ solution

Q18. One of the following compounds gives a white precipitate with dil H_2SO_4 and a green flame test. The compound is -

(A) NaCl (B) KCl

(C) $BaCl_2$ (D) $CaCl_2$

Q19. Aqueous solution of an acetate salt when treated with ferric chloride solution gives a red brown solution due to formation of :

(A) $Fe(CNS)_3$

(B) $[Fe(OH)(CH_3COO)_2]$

(C) $(CH_3COO)_2Fe$

(D) $Fe(OH_3)_2$

Q20. Al^{3+}, Cr^{3+}, Fe^{3+} are grouped together for qualitative analysis because :

(A) their carbonates are insoluble in ammonia

(B) their hydroxides are insoluble in ammonia

(C) their sulphides are insoluble in acid

(D) they belong to same group of periodic table

Q21. During group analysis reddish brown ppt. is only obtained in group III test. Which of the following inference is incorrect –

(A) Fe^{2+} may be present in the original sample

(B) Fe^{3+} may be present in the original sample

(C) (Fe^{2+} & Fe^{3+}) may be present in the original sample

(D) Fe^{3+} has to be present in the original sample

Q22. Certain yellow coloured solid gives reddish brown precipitate in group III of qualitative analysis. The solid does not react with dilute H_2SO_4 but when few drops of $KMnO_4$ solution is added to the hot suspension of salt in dilute H_2SO_4, its pink colour is discharged with effervescence and evolution of CO_2. the slat is likely to be –

(A) iron (II) oxalate (B) ferrous sulphate

(C) ferric carbonate (D) ferric chloride

Q23. An aqueous solution of $FeSO_4.Al_2(SO_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The materials obtained are
(A) a colourless filtrate and a green residue
(B) a yellow filtrate and a green residue
(C) a yellow filtrate and a brown residue
(D) a green filtrate and a brown residue

Q24. A red solid is insoluble in water. However it becomes soluble if some KI is added to water. Heating red solid in a test tube produces violet coloured fumes and droplets of metal appears on the cooler parts of test tube. The red solid is :
(A) $(NH_4)_2Cr_2O_7$ (B) HgI_2
(C) HgO (D) Pb_3O_4

Q25. A solution of KI is added to CS_2 followed by addition of Cl_2 water. A violet colour appears in CS_2 layer. If excess of Cl_2 water is added the violet colour disappears. The disappearance of violet colour is due to the formation of :
(A) I_3^- ion (B) HIO_3
(C) ICl (D) I^- ion

Q26. A yellow turbidity, sometimes appears on passing H_2S gas even in the absence of the second group radicals. This happens because –
(A) sulphur is present in the mixture as in impurity
(B) the fourth group radicals are precipitated as sulphides
(C) the H_2S is oxidised by some acidic radicals present in solution
(D) the third group radicals are precipitated

Q27. The reaction of an element A with water produces combustible gas B and an aqueous solution of C. When another substance D reacts with this solution C also produces the same gas B. D also produces the same gas even on reaction with dilute H_2SO_4 at room temperature. Element A imparts golden yellow colour to Bunsen flame. Then A , B , C and D may be identified as :
(A) Na , H_2 , NaOH and Zn
(B) K , H_2 , KOH and Zn
(C) K , H_2 , NaOH and Zn
(D) Ca , H_2 , $CaCOH_2$ and Zn

Q28. Stannous chloride reacts with excess of NaOH to form –
(A) $Sn(OH)_2$ (B) Na_2SnO_3
(C) H_2SnO_3 (D) Na_2SnO_2

Q29. Which of the following salts is insoluble in water at room temperature but dissolved in water on boiling ?
(A) $CaCl_2$ (B) $BaCl_2$
(C) $SrCl_2$ (D) $PbCl_2$

Q30. Which of the following is not preliminary test used to detect ions ?
(A) dilute H_2SO_4 test
(B) Charcoal cavity test
(C) chromyl chloride test
(D) flame test

Q31. [A] + HNO_3 (conc.) $\longrightarrow$ Black brown [B]
Red Compound
[B] + HCl $\longrightarrow$ yellow green gas [C]
[C] + $Ca(OH)_2$ $\longrightarrow$ [D]
powder
[D] + HCl $\longrightarrow$ [C]
yellow green gas
What is [D] :
(A) Cl_2 (B) $CaOCl_2$
(C) $Ca(ClO_3)_2$ (D) $CaCl_2$

Q32. Calcium burns in nitrogen to produce a white powder which dissolves in sufficient water to produce a gas (A) and an alkaline solution. The solution on exposure to air produces a thin solid layer of (B) on the surface. Identify the compounds A and B.
(A) C_2H_2, $CaCO_3$ (B) NH_3, $CaCO_3$
(C) NH_3, $Ca(OH)_2$ (D) CH_4, $CaCO_3$

Q33. A solution of $(NH_4)_2C_2O_4$, when added to a solution containing Ba^{2+}, Sr^{2+} and Ca^{2+} ions, will precipitate.
(A) Ca^{2+} ions
(B) Ca^{2+} and Sn^{2+} ions
(C) Ba^{2+}, Sr^{2+} and Ca^{2+} ions
(D) none of these

Q34. Which one among the following pairs of ions cannot be separated by H_2S in dilute hydrochloric acid ?
(A) Bi^{3+}, Sn^{4+} (B) Al^{3+}, Hg^{2+}
(C) Zn^{2+}, Cu^{2+} (D) Ni^{2+}, Cu^{2+}

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Choose the incorrect statements –
(A) Both Au and Ag dissolve in aqua regia.
(B) Using Na_2CrO_4, group-V cations (Ca^{2+}, Sr^{2+}, Ba^{2+}) are well distinguished.
(C) Using conc. H_2SO_4, Cl^-, Br^-, I^- are well distinguished.
(D) NO_2^- and NO_3^- are not distinguished by dil. H_2SO_4.

Q2. An inorganic mixture was treated with excess of dil. H_2SO_4. Effervescences were produced. The solution was heated till effervescences ceased. After this a small pinch of MnO_2 was added. Fresh effervescence was produced. Select the incorrect statements :
(A) Mixture contains CO_3^{2-} ions
(B) Mixture contains $C_2O_4^{2-}$ ions
(C) Mixture contains SO_3^{2-} ions
(D) Mixture contains both CO_3^{2-} and $C_2O_4^{2-}$ ions

Q3. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements are correct for this reaction ?
(A) Cu_2I_2 is formed
(B) CuI_2 is formed
(C) $Na_2S_2O_3$ is oxidised
(D) evolved I_2 is reduced

Q4. Which of the following pairs of ions can be separated by H_2S in dil. HCl?
(A) Bi^{3+}, Sn^{4+} (B) Al^{3+}, Hg^{2+}
(C) Zn^{2+}, Cu^{2+} (D) Ni^{2+}, Cu^{2+}

Q5. Which compound dissolve in hot dil. HNO_3 ?
(A) HgS (B) PbS
(C) CuS (D) CdS

Q6. Which of the following statement(s) is/are correct with reference to the ferrous and ferric ions ?
(A) Fe^{3+} gives brown colour with potassium ferrocyanide
(B) Fe^{2+} gives blue precipitate with potassium ferrocyanide
(C) Fe^{3+} gives red colour with potassium thiocyanate
(D) Fe^{2+} gives brown colour with ammonium thiocyanate

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 7 to 9

Pyrolusite on heating with KOH in the presence of air gives a dark green compound (A). The solution of (A) on treatment with H_2SO_4 gives a purple coloured compound (B), which gives following is reactions:
(i) KI on reaction with alkaline solution of (B) changes into a compound (C)
(ii) The colour of compound (B) disappears on treatment with the acidic solution of $FeSO_4$.
(iii) With conc. H_2SO_4 compound (B) gives (D) which can decompose to yield (E) and oxygen.

Q7. The compound (C) is
(A) I_2 (B) I_2O_5
(C) KIO_3 (D) KIO_4

Q8. The comopund (E) is:
(A) MnO (B) MnO_2
(C) Mn_2O_3 (D) Mn_3O_4

Q9. Oxidation state of manganese of the compound (A) is -

(A) +2 (B) +4

(C) +7 (D) +6

Paragraph for Q. 10 to 12

(i) Salt 'A' $\xrightarrow{\text{heated}}$ Anhydrous.

'A' $\xrightarrow[\text{heated}]{\text{strongly}}$ B + C

(ii) Aqueous solution of

$C + Hg(NO_3)_2$ solution $\rightarrow$ D$\downarrow$ (yellow ppt.)

(iii) Aqueous solution of

$A + KCN \rightarrow$ [colourless solutin] $\xrightarrow[\text{solution}]{FeCl_3}$ Blood red colouration.

Q10. The ppt. D is –

(A) HgO (B) $HgO.HgSO_4$

(C) $HgSO_4.2HgO$ (D) CdS

Q11. The structure of anionic part of B is –

(A) linear (B) Zigzag chain

(C) planar ring (D) Nonplanar ring

Q12. The number of water of crystallisation in 'A' is –

(A) 4 (B) 3

(C) 2 (D) 5

Paragraph for Q. 13 to 15

(i) A certain inorganic compounds (X) shows the following reactions on passing H_2S through acidified solution of (X) a brown ppt is obtained.

(ii) The ppt. obtained in (i) is dissolved in excess of yellow $(NH_4)_2S_2$.

(iii) On adding an aqueous solution of NaOH of solution of (X) first white ppt. is obtained, which dissolves in excess of NaOH.

(iv) The compound X reduces $FeCl_3$ solution.

Q13. The soluble complex which is obtained with excess of NaOH on oxidation produces -

(A) Na_2SnO_3 (B) Na_2MnO_4

(C) Na_2ZnO_2 (D) Na_2AlO_2

Q14. When salt X reacts with $AuCl_3$ we get –

(A) purple of cassius (B) liquid gold

(C) gold rush (D) candy fluid

Q15. The salt (X) gives grey mass and another compound (Z) with corrosive sublimate. (Z) on reaction with NH_4Cl gives (Y). Which is the correct statement about (Y).

(i) it is known as pink salt

(ii) it is used as mordant

(iii) it is used as moderator in nuclear reactors

(iv) it is known as scheele's salt

(A) i, ii (B) i, iii

(C) ii, iv (D) iii, iv

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q16. Match the column –

Column I **Salt**	**Column II** **Production on heating**
(A) $NaNH_4HPO_4$	(P) NH_3
(B) $Na(NH_4)_2PO_4$	(Q) $NaPO_3$
(C) NaH_2PO_2	(R) $Na_2P_4O_7$
(D) $Na_2NH_4PO_4$	(S) PH_3

Q17. Column I contains compounds which on heating gives the compounds in Column II

Column I	**Column II**
(A) $(NH_4)_2Cr_2O_7$	(P) NO_2
(B) $K_2Cr_2O_7$	(Q) Cr_2O_3
(C) NH_4NO_2	(R) O_2
(D) $AgNO_3$	(S) N_2

Q18. Match the column –

Column I	**Column II**
(A) NaCl	(P) Soluble in water
(B) AgCl	(Q) Lattice energy > Hydration energy
(C) $CdCl_2$	(R) Lattice energy < Hydration energy
(D) $FeSO_4$	(S) Cation has Pseudo inert (18 e^- in valence shell) gas configuration

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q19. **Statement - 1 :** Sulphate is estimated as $BaSO_4$ and not as $MgSO_4$.
Statement - 2 : Ionic radius of Mg^{2+} is smaller than that of Ba^{2+}.

Q20. **Statement 1 :** To convert Mn^{2+} into MnO_4^-, instead of (PbO_2 + conc. HNO_3), (Pb_3O_4 + conc.HNO_3) can also be used.
Statement 2 : PbO is soluble in cocn. HNO_3 while PbO_2 is not soluble in conc. HNO_3.

Q21. **Statement 1 :** Cu^{2+} and Cd^{2+} are separated from one another by first adding KCN solution and then passing H_2S gas.
Statement 2 : KCN reduces Cu^{2+} to Cu^+ and forms a complex with it.

Q22. **Statement - 1 :** Aqueous solution of $FeCl_3$ is acidic.
Statement - 2 : The hydrated form of $FeCl_3$ consist of six H_2O molecules as, $FeCl_3 \cdot 6H_2O$.

Q23. **Statement - 1 :** Conc. H_2SO_4 reacts with KCl to give Cl_2 gas.
Statement - 2 : HCl can not be oxidized by conc. H_2SO_4.

Q24. **Statement - 1 :** The solubility of sulphates decreases on moving down the 2nd group of modern periodic table.
Statement - 2 : The solubility product values of sulphates increases on moving down the 2nd group of modern periodic table.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	B	D	A	C	B	D	C	D	B
11	12	13	14	15	16	17	18	19	20
C	A	C	C	B	B	A	D	C	D
21	22	23	24	25	26	27	28	29	30
B	D	A	A	A	B	A	C	B	B
31	32	33	34	35	36	37			
B	B	B	B	A	C	B			

Ex-1A

1. A, C, D **2.** A, B, D **3.** A, C **4.** A, B, C, D **5.** A, C, D
6. A, B **7.** B, C, D **8.** A, B, D **9.** A **10.** B
11. A **12.** C **13.** C **14.** B **15.** C
16. A **17.** B
18. A → P, S ; B → S ; C → Q ; D → R
19. A → S ; B → S ; C → R ; D → P, Q
20. A → R ; B → S ; C → Q
21. A → Q, R ; B → R ; C → P, Q
22. D **23.** A **24.** A **25.** A **26.** A
27. A **28.** C **29.** C **30.** A

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	B	A	C	A	D	D	C	B
11	12	13	14	15	16	17	18	19	20
D	C	B	B	D	A	B	C	B	B
21	22	23	24	25	26	27	28	29	30
D	A	C	B	B	C	A	D	D	C
31	32	33	34						
B	B	C	A						

Ex-2A

1. A, B, D **2.** A, B, C **3.** A, C, D **4.** B, C, D **5.** B, C, D
6. B, C **7.** C **8.** B **9.** D **10.** C
11. B **12.** D **13.** A **14.** A **15.** B
16. A → P, Q; B → P, Q; C → P, Q; D → P, R
17. A → Q, S; B → Q, R; C → S; D → P, R
18. A → P, R; B → Q, S; C → P, R S; D → P, R
19. B **20.** A **21.** B **22.** B **23.** D
24. C